

■ JavaScript Operator Real-Life Problem: E-Commerce Discount System

Problem Statement:

You are building a checkout feature for an e-commerce app. Your goal is to calculate the final payable amount based on multiple discount conditions, using JavaScript operators.

Requirements:

- 1 If the user enters a coupon code (e.g., 'SAVE20'), extract the discount percentage from it using `slice()` and convert it to a number using `parseInt()`.
- 2 Apply that discount percentage to the cart total.
- 3 If the user is a member, apply an additional 10% discount.
- 4 If the final amount after discounts is greater than or equal to ■500, delivery is free. Otherwise, add ■50 as delivery charge.
- 5 Use a ternary operator to display: 'Free delivery applied!' or 'Delivery charge ■50 applied.'
- 6 Use arithmetic, comparison, logical, and assignment operators for all calculations.
- 7 Convert the final payable amount into a string prefixed with the '■' symbol using `toString()`.

■ Bonus Challenges:

- If the user enters 'SAVE50', apply a 50% discount.
- Handle cases where cart total is 0 → print 'Invalid cart amount' using logical and ternary operators.
- Show discount applied, membership discount, and final payable amount neatly formatted in the console.

■ Example Output:

Original total: ■550
Discount applied: ■110
Membership discount: ■44
Free delivery applied!
Final amount to pay: ■396

■ *Tip: Try to write this logic step by step using operators only — no libraries. This exercise will help you revise all operator concepts like arithmetic, logical, comparison, assignment, and ternary operators.*